

Emma K. Brown
PhD Candidate

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EDUCATION

- 2021 — present **Arizona State University**, School of Earth & Space Exploration
Geological Sciences, *PhD Candidate*
Dissertation: Thermodynamic, Kinetic and Biological Perspectives on Transformations of Carbon in Natural Systems (*in prep.*)
Committee: Drs. Everett Shock (Advisor; ASU), Elizabeth Trembath-Reichert (ASU), Ian Gould (ASU), Hilairy Hartnett (UW), Julie Castillo-Rogez (ASU)
- 2018 — 2021 **Georgia Institute of Technology**, School of Earth & Atmospheric Sciences
B.S. in Earth & Atmospheric Sciences; Minor in Physics; *high honors*
Undergraduate Research Project: The Geochemistry and Biology of Hyper-saline Lake Environments
Project Advisors: Drs. Britney Schmidt and Jacob Buffo

RESEARCH EXPERIENCE

- Arizona State University**, School of Earth & Space Exploration
2021 — present *Graduate Research Associate* HOG, GEOPIG, and MOD Lab Groups
- Studying the effects of aqueous metals and minerals on organic acid decomposition through hydrothermal experiments and geochemical modeling
 - Building a predictive framework for the kinetics of organic acid decarboxylation
 - Developing a database and estimation procedures for the thermodynamic properties of aqueous aromatic carboxylic and polyfunctional acids and their metal complexes across temperature
 - Characterizing heterotrophic inorganic carbon fixation in marine and soil isolates across various culture conditions using single-cell and bulk stable isotope probing
- Georgia Institute of Technology**, School of Earth & Atmospheric Sciences
2019 — 2021 *Undergraduate Research Associate* PH&T Lab Group
- Studied the geochemistry and microbiology of endorheic hypersaline lakes to characterize the habitability of extreme environments through fieldwork, labwork, and modeling

TEACHING & MENTORSHIP

- Arizona State University**
- 2024 — 2026 *Graduate Peer Mentor* for first-year graduate students in the department
- 2024 — 2025 *Mentor* for undergraduate NASA/Arizona Space Grant Consortium research project
- 2021 — 2022 *Sundial Mentor* for summer camp & semester classes for incoming undergraduates
- Spring 2022 *Instructor* for three sections of Introduction to Geology Lab I
- Geoscience Education and Mentorship Support (GEMS)**
- 2022 — 2023 *Graduate Student Mentor* for undergraduates and those applying to graduate school
- Georgia Institute of Technology**
- Fall 2020 *Head Teaching Assistant* for Earth Processes Lab
- 2019 — 2020 *Teaching Assistant* for Earth Processes Lab
- 2020 — 2021 *Vice President* for GT chapter of National Honors Society for the Earth Sciences
- Communities in School**
- 2017 — 2018 *Volunteer Tutor* for Middle School Algebra and High School Environmental Science

AFFILIATIONS & POSITIONS

Network for Ocean Worlds Research Coordination Network

2026 — present *Social & Conference Chair* for the Future Leader of Ocean Worlds

Europa Clipper Project Science

2022 — present *Graduate Student Affiliate* on the MASPEX team

AWARDS

Spring 2027 Focus on Finishing Your Degree Fellowship, Arizona State University (\$15,000)
Spring 2027 University Graduate Fellowship, Arizona State University (\$1,838)
2023 — 2026 NASA Future Investigator (FINESST) (\$149,421)
Summer 2023 Summer Exploratory Graduate Fellowship, Arizona State University (\$8,293)

RESEARCH EXPEDITIONS

Yellowstone National Park

July 2025 Fieldwork assistant for geochemical, biological and gas sampling of hot springs
July 2023 Fieldwork assistant for stable isotope probing experiments in hot springs
July 2022 Fieldwork assistant for hot spring incubation experiments to detect novel microbes

Kama'ehuakanaloa Seamount *on the R/V Kilo Moana*

Jan. 2023 Hydrothermal vent fluid sampling and stable isotope probing experiments

Cariboo Plateau, British Columbia, Canada

Sept. 2023 Fieldwork assistant for the biogeochemical sampling of hypersaline lakes
Feb. 2020 Fieldwork assistant for sampling icy hypersaline lakes
Sept. 2019 Fieldwork assistant for the biogeochemical sampling of hypersaline lakes

INVITED PRESENTATIONS

Bose Lab Group Meeting, Arizona State University. The Kinetics and Thermodynamics of Organic Acid Decomposition in Hydrothermal Systems. Nov. 20th, 2026.

SESE Department Colloquium, Arizona State University. What Does Organic Chemistry Look Like on Ocean Worlds in the Absence of Life? Jan. 14th, 2026.

Dekas Lab Group Meeting, Stanford University (virtual). Heterotrophic Inorganic Carbon Fixation in Different Growth Conditions Using Stable Isotope Probing. Dec. 8th, 2025

Life Detection Discussion Hour, Arizona State University. Effect of Salt on Organic Preservation. Oct. 7th, 2024.

Exploring Ocean Worlds ICAR Group Meeting (virtual). How Abiotic Organic Transformations Can Constrain Biosignatures on Ocean Worlds. April 24th, 2024.

CONFERENCE PRESENTATIONS

Brown, E. K., Gould, I., Hartnett, H., Shock, E., (2026) Looks Like Lunch: Persistence of Organic Acids in Salty Solutions. **AbGradCon**, Tucson, AZ., Sept. 15th. (Talk)

Brown, E. K., Gould, I., Hartnett, H., Shock, E., (2026) Salty Solutions: Hydrothermal Organic Decomposition Promotes Mineral Formation in Ocean World Conditions. **AbSciCon**, Madison, WI., May 18th. (Talk)

Brown, E. K., Gould, I., Hartnett, H., Shock, E., (2026) What Does Organic Chemistry Look Like on Ocean Worlds in the Absence of Life? **Arizona Astrobiology Symposium**, Tucson, AZ., March 28th. (Poster)

Brown, E. K., Pulliam, K. P., Trembath-Reichert, E. (2025) Microbial response to environmental conditions: heterotrophic inorganic carbon fixation and morphological alterations.

Arizona-Nevada Branch of ASM's Annual Conference, Flagstaff, AZ., April 26th. (Poster)

Brown, E. K., Gould, I., Hartnett, H., Shock, E. (2025) The Fate of Biosignatures: Salty Solutions Slow Down Decarboxylation. **Exploring the Abiotic Background for Life Detection Workshop**, Washington, D.C., March 31st. (Poster)

Brown, E. K., Gould, I., Hartnett, H., Shock, E., (2024) How Abiotic Organic Transformations Can Constrain Biosignatures on Ocean Worlds. **AbSciCon**, Providence, RI., May 10th. (Talk)

Brown, E. K., Buffo, J., Schmidt, B., (2021) Trapped in the ice: An analysis of brines in British Columbia's hypersaline lakes. **Lunar and Planetary Science Conference**. (Virtual Poster)

PUBLICATIONS

Brown, E. K., Pulliam, K. P., Harris, C., Weeks, K., Leavitt, W., Saunders, J., Trembath—Reichert, E. (2026) Heterotrophic inorganic carbon fixation by two common Gammaproteobacteria varies in response to temperature and resource availability. *Applied and Environmental Microbiology*, e00628-26.

Brown, E. K. and Shock, E., (*in prep for Geochim. Cosmochim. Acta*) Thermodynamic properties of aqueous aromatic compounds and their metal complexes across temperature and pressure.

Brown, E. K., Gould, I., Hartnett, H., Shock, E., (*in prep for ACS Earth and Space Chemistry*) The role of metal-organic complexation in the hydrothermal decomposition of phenylacetate: reaction kinetics and carbonate formation.

Davila, A., Hoehler, T., Shkolyar, S., Nuevo, M., Rogers, K., Barge, L. M., ... & **Workshop Participants**. (2026). The Abiotic Background: Challenges and Opportunities in the Search for Evidence of Life Beyond Earth. *Astrobiology*, 15311074261484864.

Buffo, J. J., **Brown, E. K.**, Pontefract, A., Schmidt, B. E., Klempay, B., Lawrence, J., ... & OAST Team. (2022). The bioburden and ionic composition of hypersaline lake ices: Novel habitats on Earth and their astrobiological implications. *Astrobiology*, 22(8), 962-980.

TECHNIQUES & SKILLS

Gas Chromatography - Flame Ionization Detection, Gas Chromatography - Mass Spectrometry, Fourier Transform Infrared Spectroscopy, Nuclear Magnetic Resonance, Elemental Analysis Isotope Ratio Mass Spectrometry, Spectrophotometry, Raman Microscopy, Fluorescent Microscopy, Ion Chromatography, Single-cell and bulk Stable Isotope Probing, Hydrothermal Experiments (high temperatures and pressures, vacuum systems, sealing ampules, etc.), Organic Chemistry Wet-Lab (synthesizing simple compounds, derivatizations and extractions), Biology Wet-Lab (culture maintenance, cell counts, growth curves, biomolecule extractions, freeze-drying), Computational (python, MATLAB), Thermodynamic and Geochemical Modeling (EQ3/6, CHNOSZ), Adobe Illustrator

PROFESSIONAL EXPERIENCE

2020 — 2021	Warehouse Associate, United Parcel Service
2020	Temp. Certified Nursing Assistant, Cypress Skilled Nursing
2019	Department Student Assistant, Georgia Institute of Technology
2018	Nursing Research Intern, Children's Healthcare of Atlanta
2016 — 2017	Editor-in-Chief, Johns Creek High School Newspaper
2013 — 2017	Customer Service Clerk, Publix Super Markets